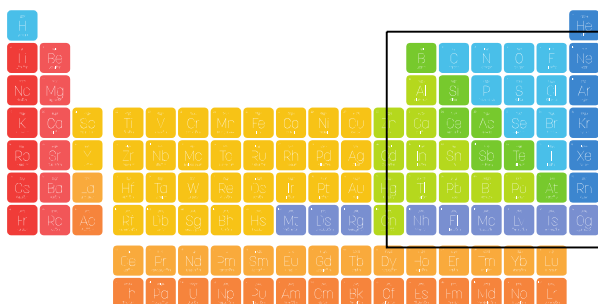


JEE (MAIN+ADVANCED)

Organic Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Carbonyl Compounds

English Medium

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03
Carbonyl Compounds

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[illegible]

Carbonyl Compounds

1.0 Classification And General Introduction

Organic Compounds having $>C=O$ group are called carbonyl compounds and $>C=O$ group is known as carbonyl or oxo group. Its general formula is $C_n H_{2n} O$ ($n = 1, 2, 3, \dots$). Carbonyl compounds are grouped into two categories.

(a) **Aldehydes** : Aldehyde group is $\overset{\overset{O}{\parallel}}{>C}-H$ (also known as formyl group). It is a monovalent group

(b) **Ketones** : The carbonyl group ($>C=O$) is a Ketonic group when its both the valencies are satisfied by alkyl group. It is a bivalent group.

Structure

In $>C=O$ compounds C-atom is sp^2 hybridised which forms two σ bonds with C and H-atom respectively and one σ bond with oxygen atom. The unhybridised atomic orbital of C-atom and the parallel 2p orbital of oxygen atom give the π bond in $>C=O$ group.

Due to electro-negativity difference in C & O atoms, the $>C=O$ group is polar.

$\overset{\delta+}{>C}=\overset{\delta-}{O}$ Hence aldehydes and Ketones possess dipole moment.

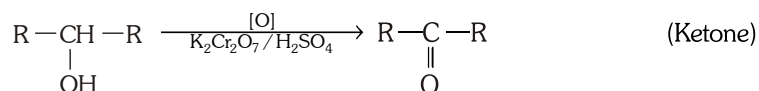
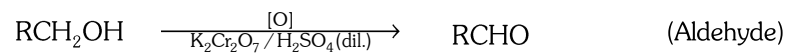
GENERAL METHODS OF PREPARATION

2.0 FOR BOTH ALDEHYDES AND KETONES

2.1 By Oxidation (or dehydrogenation) of Alcohols

(a) By $K_2Cr_2O_7 / H_2SO_4$:

Oxidation of primary alcohols gives aldehyde and oxidation of secondary alcohols gives Ketones. Here, $(K_2Cr_2O_7 / H_2SO_4)$ is a strong oxidising agent.



Aldehydes are quite susceptible to further oxidation to acids -

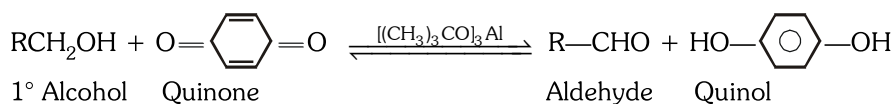
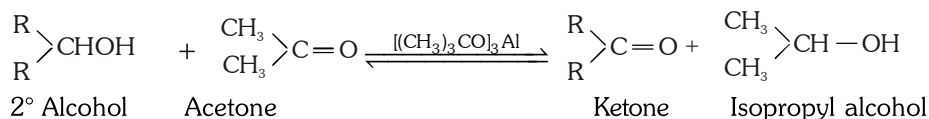


Thus oxidation of primary alcohols is made at the temperature much above the boiling point of aldehyde and thus aldehydes are vapourised out and prevented from being oxidised.

Note : Aldehydes can be prepared from 1° alcohol, secondary alcohols can be oxidized to ketones, by oxidation with pyridinium chlorochromate (PCC) in CH_2Cl_2 solvent, pyridinium dichromate (PDC) and with Collin's reagent (CrO_3 , pyridine/cold)

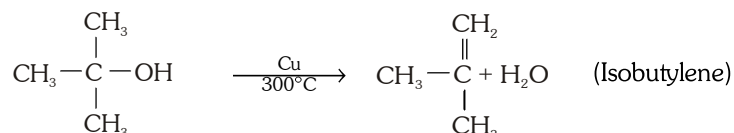
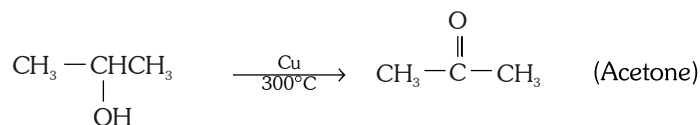
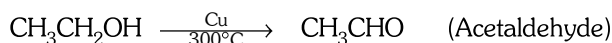
(b) Oppenauer Oxidation

The oxidation of secondary alcohols to ketones by heating them with specific reagent : $[(CH_3)_3CO]_3Al$ (Aluminium-t-butoxide) in presence of acetone. Primary alcohols may also be oxidized to aldehydes; if ketone is replaced by a better hydrogen acceptor, (e.g. p-benzoquinone), the equilibrium can be controlled by the amount of acetone, an excess of which favours the oxidation of the alcohol.



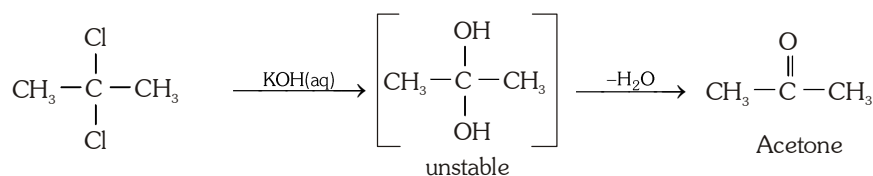
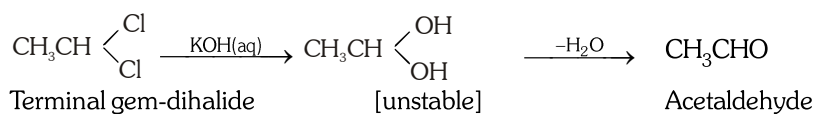
Note : The reaction is the reverse of Meerwein-Ponndorf-verley reduction.

(c) By Cu/300°C



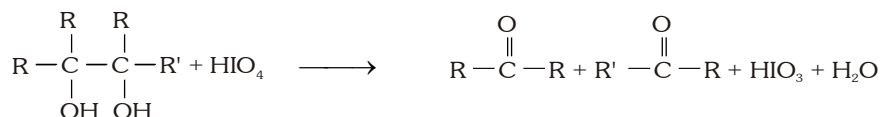
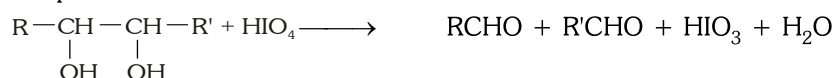
2.2 Preparation Of Carbonyl Compound By Alkaline Hydrolysis Of Geminal Dihalide

Terminal gem-dihalides on hydrolysis give aldehydes while the non-terminal dihalides give ketone.



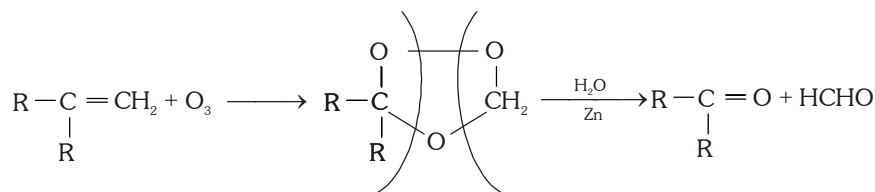
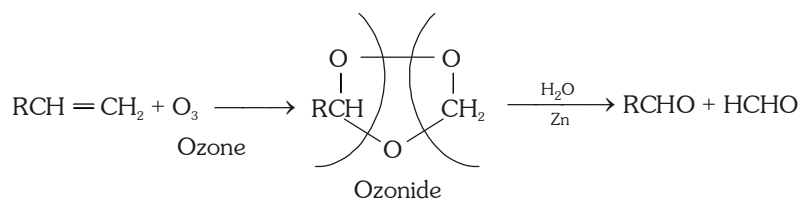
2.3 By Oxidation of diols

With periodic acid (HIO_4) or lead tetra acetate $[(\text{CH}_3\text{COO})_4\text{Pb}]$ vicinal diols gets oxidised to form carbonyl compounds



2.4 By Ozonolysis of alkenes

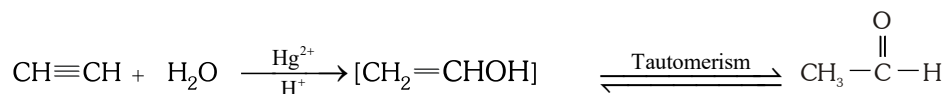
This reaction is used to determine the position of double bond in alkene.



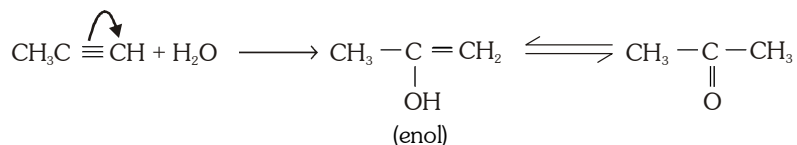
2.5 From Alkyne

(a) Preparations Of Carbonyl Compounds From Hydrolysis Of Alkyne (Kucherov Reaction):

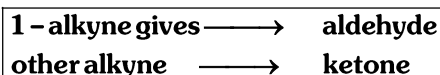
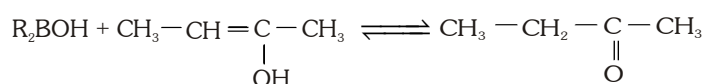
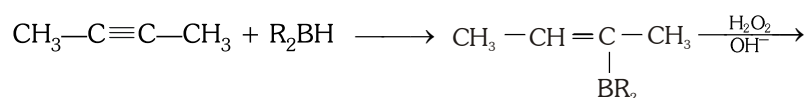
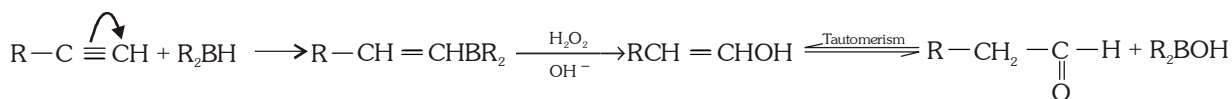
With dil H_2SO_4 & 1% HgSO_4 at $60-80^\circ\text{C}$.



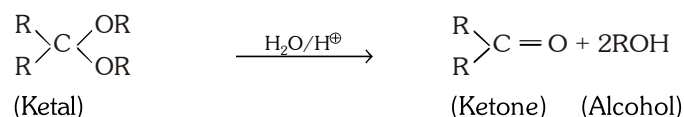
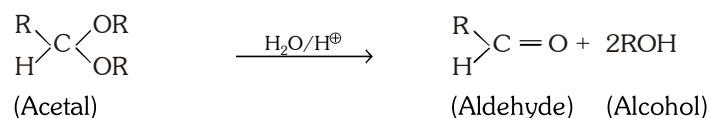
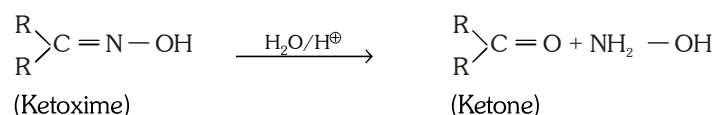
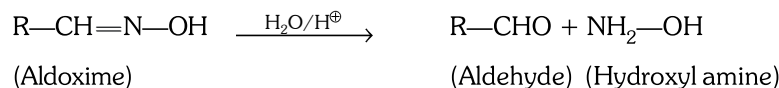
• Other alkynes give ketone



(b) Hydroboration-oxidation : Reaction with B_2H_6 , 2BH_3 or R_2BH give dialkyl borane.

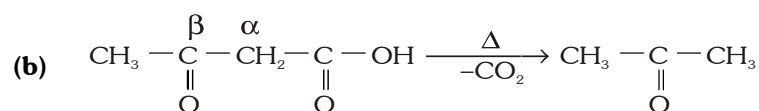
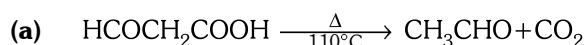


2.6 By hydrolysis of carbonyl derivatives



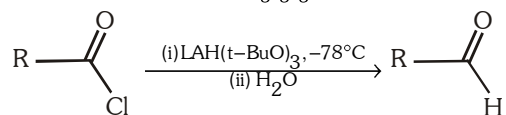
2.7 From β -keto acids

The decarboxylation reaction takes place via formation of six membered ring transition state.

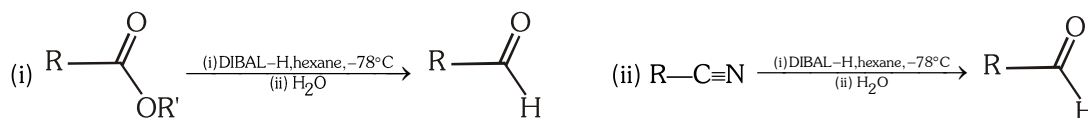


3.1 Reduction of acyl halides, esters and nitriles

Acyl chlorides, esters and nitriles can be reduced to aldehydes by treating them with lithium-tri-tert butoxyaluminium hydride, $\text{LiAlH}[\text{OC}(\text{CH}_3)_3]_3$ and DIBALH (di iso propyl aluminium hydride) at -78°C .

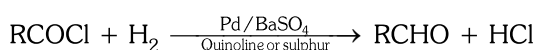


Both esters and nitriles can be reduced to aldehydes by DIBAL-H. Reduction must be carried out at low temperatures. Hydrolysis of the intermediates gives the aldehyde.

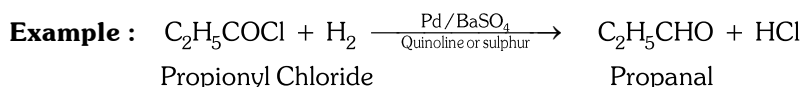


3.2 Rosenmund's reduction

Quinoline or sulphur act as a poisoned catalyst, controls the further reduction of aldehyde to alcohols.

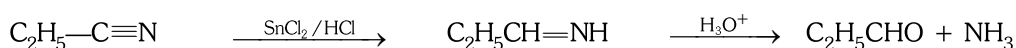
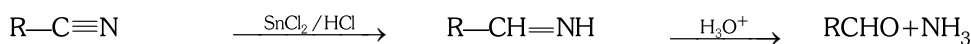


Formaldehyde can not be prepared by this method.

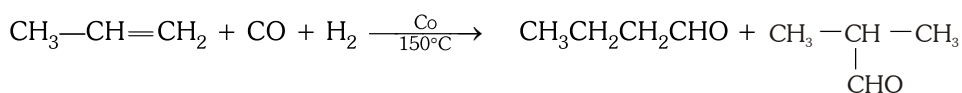


3.3 Stephen's reduction

Alkyl cyanides are reduced by SnCl_2 and HCl .



3.4 Oxo reaction or hydroformylation

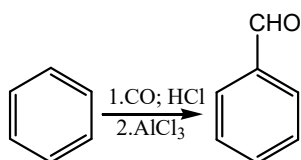


Note : In this reaction symmetrical alkene gives 1 aldehyde while unsymmetrical alkene gives isomeric aldehyde (Chain isomers).

3.5 Preparation of Aromatic Aldehydes

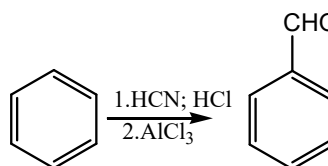
Gattermann-Koch Reaction

1. Gattermann-Koch aldehyde synthesis



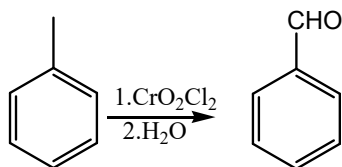
Gattermann-Aldehyde Synthesis

2. Gattermann aldehyde synthesis



Etard's Reaction

3. Etard's reaction



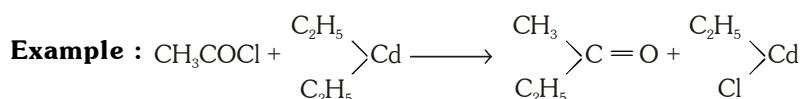
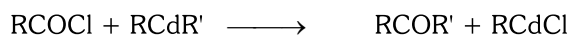
4.0 FOR KETONES ONLY

4.1 From Grignard's reagent



4.2 Preparation of Ketone From Acyl Chloride (dialkyl Cadmium)

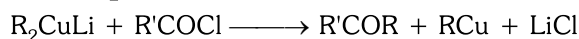
RCdR' (dialkyl Cadmium) is an organometallic compound.



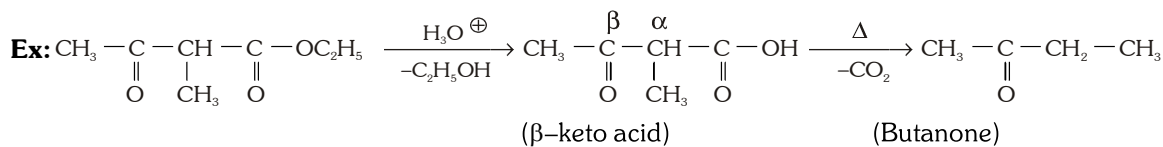
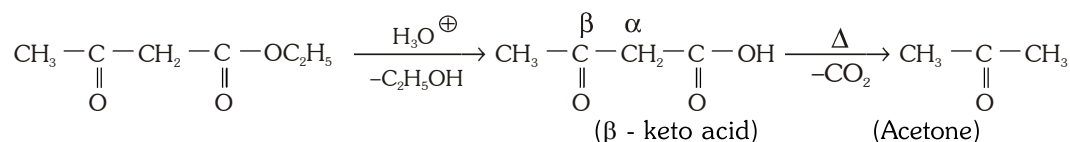
Note:

This reaction is superior than Grignard Reaction because the ketones formed, further reacts with Grignard reagent to form 3° alcohols.

4.3 From R_2CuLi

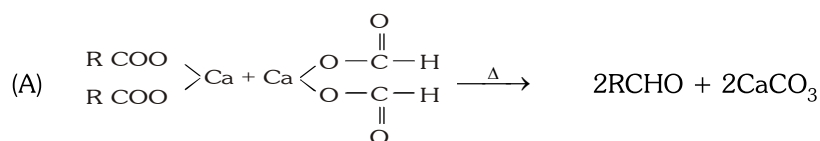


4.4 By hydrolysis of Aceto Acetic Ester (AAE)

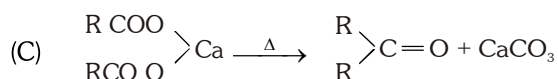
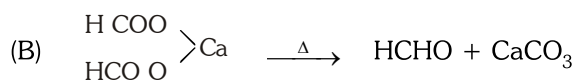


5.0 OTHER METHODS FOR ALDEHYDE AND KETONE

5.1 From dry distillation of Calcium and Barium Carboxylate Salts



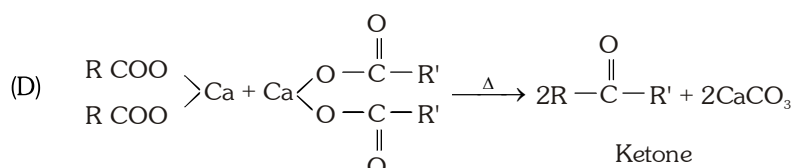
Calcium formate (Also $\text{R}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{R}$ and HCHO formed)



Calcium-alkanoate

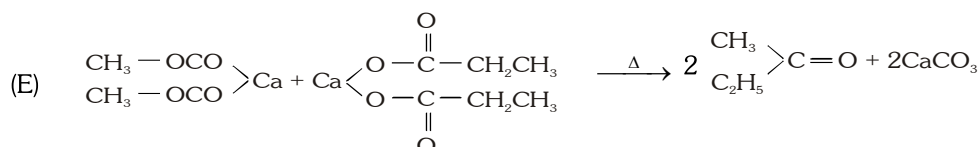
Ketone

Calcium salts of acids other than formic acid on heating together gives ketone



Ketone

To prepare ethyl methyl ketone Calcium acetate and Calcium propionate are used :



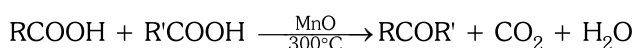
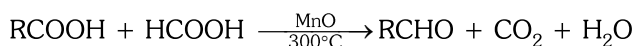
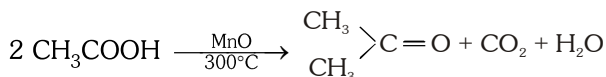
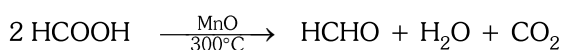
Calcium Acetate

Calcium propionate

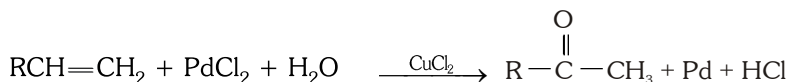
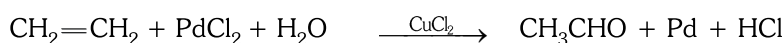
Ethyl methyl ketone

5.2 By Thermal decomposition of carboxylic acids (By heating with MnO)

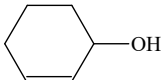
Vapour of carboxylic acids when passed over $\text{MnO}/300^\circ\text{C}$ give carbonyl compounds

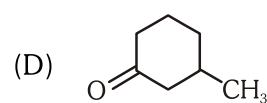
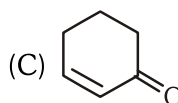
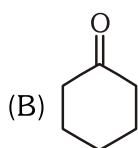
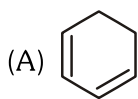


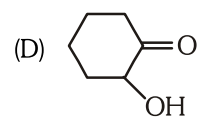
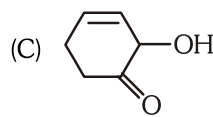
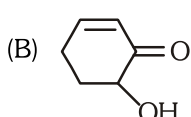
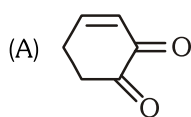
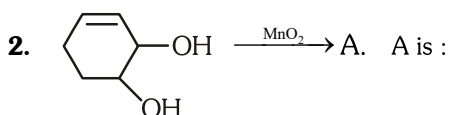
5.3 Wacker Process



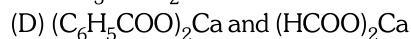
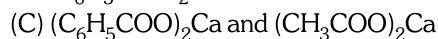
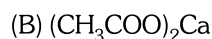
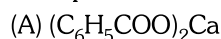
BEGINNER'S BOX-1

1.  $\xrightarrow[\text{acetone}]{\text{Aluminium tert. butoxide}}$ A in the above reaction A is :





3. Acetophenone can be obtained by the distillation of:



4. Gem dihalide on hydrolysis gives:

(A) Vic diol

(B) Gem diol

(C) Carbonyl compound

(D) Carboxylic acid

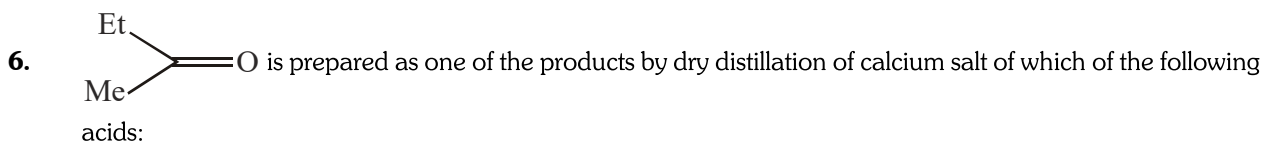
5. Acetal or ketal is:

(A) Vic dialkoxy compound

(B) α, ω -dialkoxy compound

(C) α -alkoxy alcohol

(D) Gem dialkoxy compound

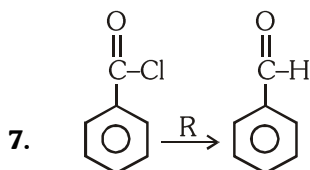


(A) ethanoic acid and methanoic acid

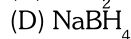
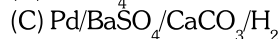
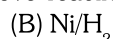
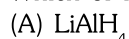
(B) Propanoic acid and methanoic acid

(C) Propanoic acid and ethanoic acid

(D) None of these



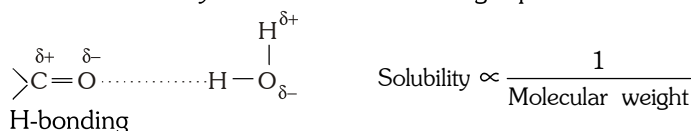
Which of the following can be used as R in the above reaction :



6.0 Physical Properties of Carbonyl Compounds :

• Solubility

Aldehyde and ketones having upto 3 carbons (formaldehyde, acetaldehyde and propionaldehyde and acetone) are quite soluble in water due to polarity of $\text{>C}=\text{O}$ bond and can form H—bond with water molecule. Aldehydes and ketones having equal to or more than five carbons are insoluble in water.



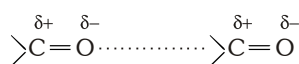
H-bonding

• Boiling point

Boiling point $\propto$ Molecular weight

Boiling point order is - **Alcohol** > **Carbonyl compounds** > **Alkane**

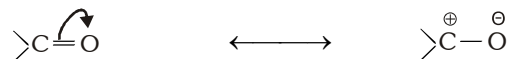
This is because in alcohols intermolecular H-bonding is present but in carbonyl compounds H-bonding doesn't exist, instead dipole-dipole attraction and vander waal force of attraction is present. Alkanes are non polar.



7.0 CHEMICAL PROPERTIES

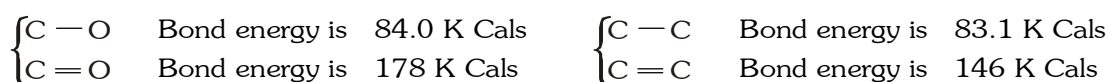
7.1 Nucleophilic of Carbonyl Compounds

Due to strong electronegativity of oxygen, the mobile π electrons pulled strongly towards oxygen, leaving the carbon atom deficient of electrons.



Carbon is thus readily attacked by $\ddot{\text{N}}\text{u}$. The negatively charged oxygen is attacked by electron deficient (electrophile) E^+ .

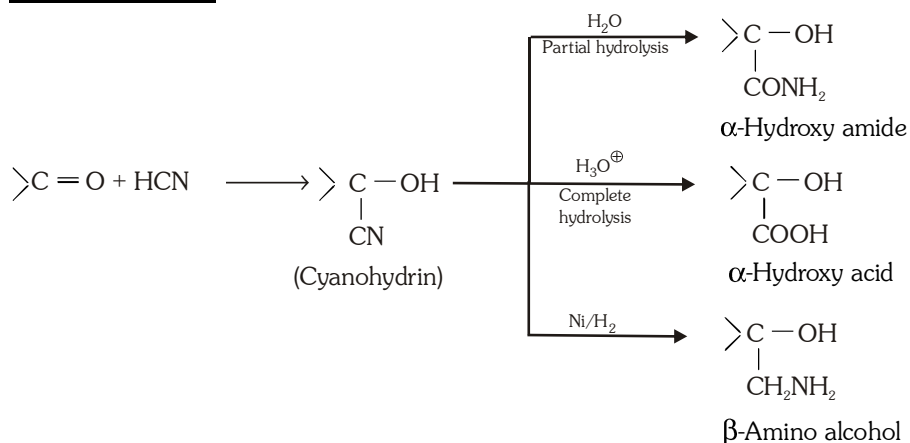
$\text{>C}=\text{O}$ bond in carbonyl group is stronger than $\text{C}=\text{C}$ bond in alkanes.



(Reactivity of carbonyl group $\propto$ Magnitude of +ve charge)

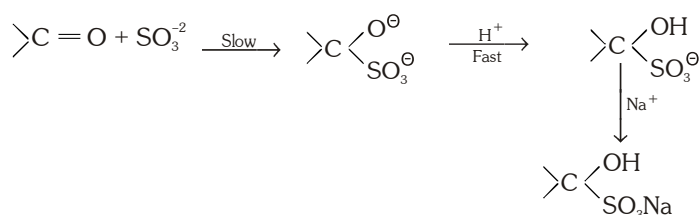
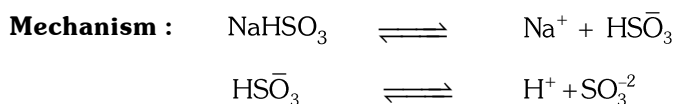
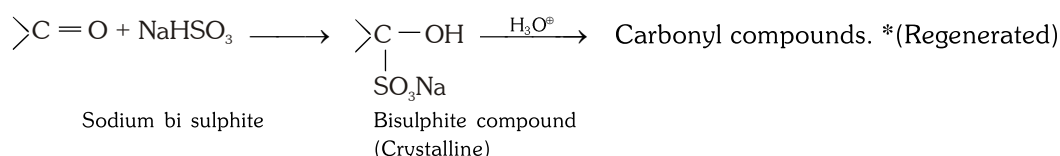
7.2 Nucleophilic Addition Reactions

• Addition of HCN



• Addition of NaHSO_3

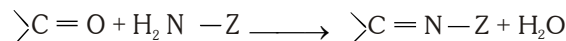
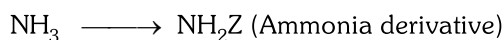
This reaction is utilized for the separation of carbonyl compounds from non-carbonyl compounds.



Only Aldehyde and Methyl Ketones can give this reaction.

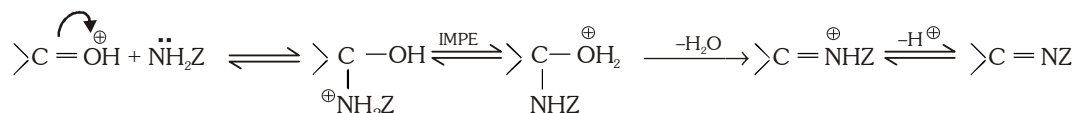
• **Reaction with ammonia derivatives**

These are condensation or addition elimination reaction. These reactions proceed well in weakly acidic medium.



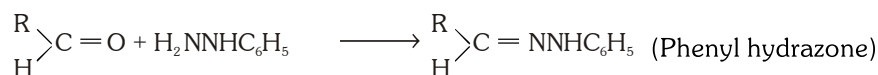
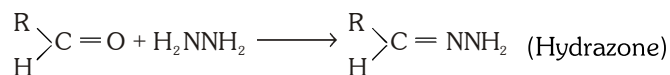
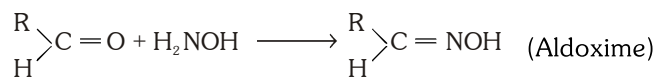
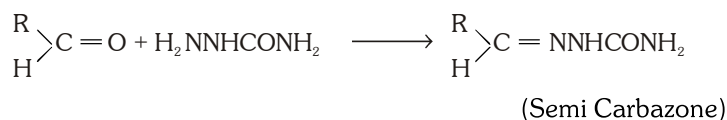
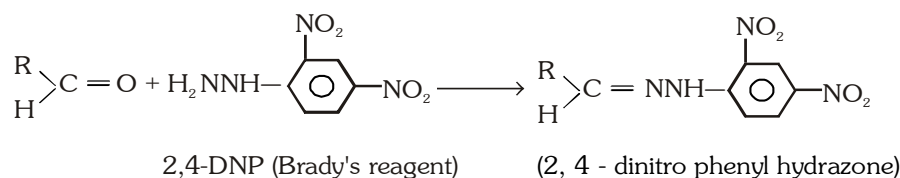
Addition - elimination (Condensation)

• **Mechanism**

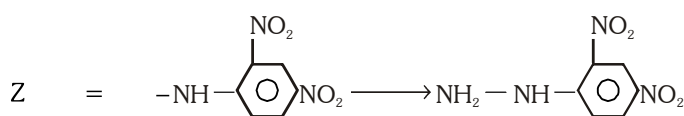
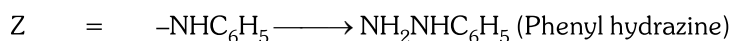
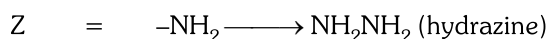
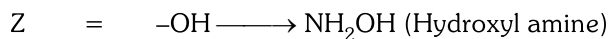


• **Reactions of Carbonyl group with Ammonia derivatives (Z)**

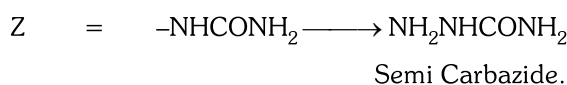
2, 4-DNP (Brady's reagent)



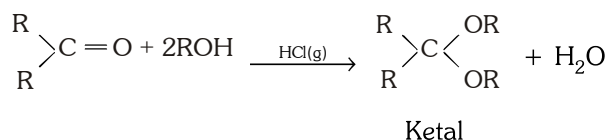
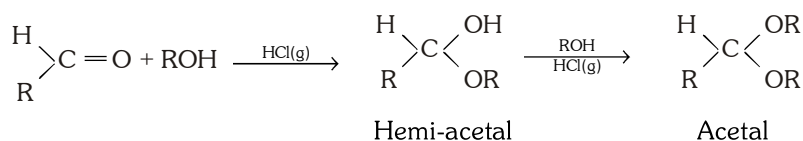
• **Ammonia derivatives (NH₂Z)**



2, 4-Dinitro phenyl hydrazine (DNP) Brady's reagent.

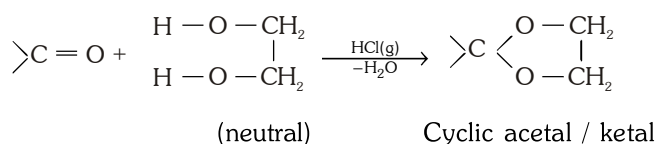


• **Nucleophilic Addition of Alcohol**

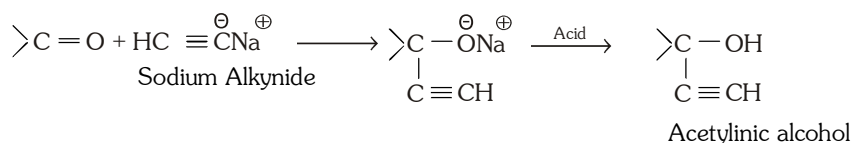


Note : Acetal / ketal is formed only in acidic medium. Hemiacetal is formed in acidic as well as basic medium. As a result, acetals are stable in basic medium and unstable in acidic medium.

• **Reaction with glycol (group protection)**

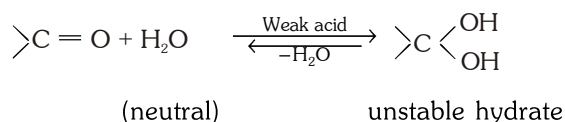


• **Reaction with sodium alkynide**



• **Nucleophilic Addition of H₂O**

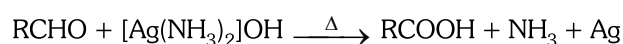
It is a reversible reaction.



HCHO, chloral, hexafluoroacetone are some of the compounds which form stable hydrates.

4.0 OXIDATION OF CARBONYL COMPOUNDS

1. Tollen's Reagent



Aldehyde acts as reducing agent, they can reduce mild oxidizing agents like Tollen's Reagent. Tollen's test Gentle Heating for 20 to 25 mins.

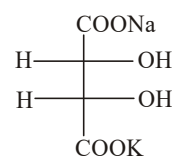
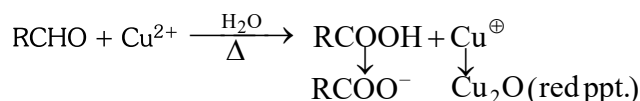
2. Fehling's Solutions

Fehling's A
aq. CuSO₄

Fehling's B
Alk. solution of Roschelye
salt (sodium potassium tartrate)

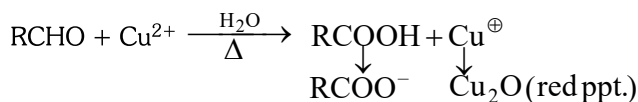
It act's a carrier for Cu²⁺ as it make reversible complex with Cu²⁺

This test is also used is Blood and Urine test.

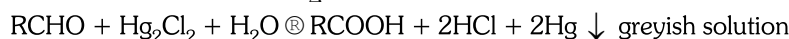
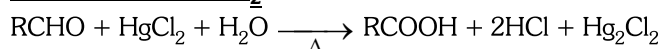


3. Benedict's Test

Sodium Citrate + NaOH + NaHCO₃ + CuSO₄



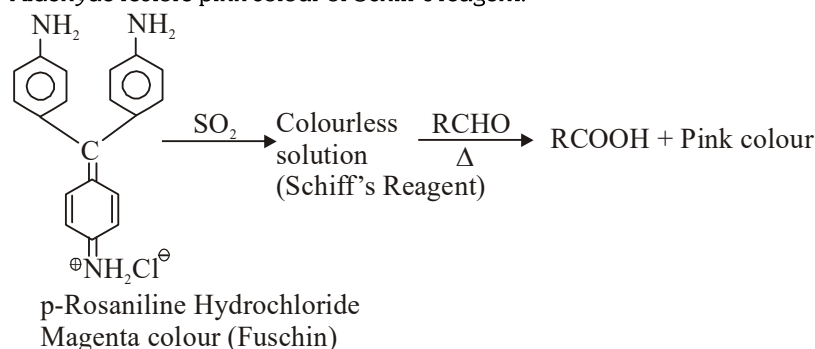
4. Reaction with HgCl₂



5. Schiff's Reagent

Schiff's Reagent is aq. solution of following base decolourised by passing SO_2 .

Aldehyde restore pink colour of Schiff's reagent.

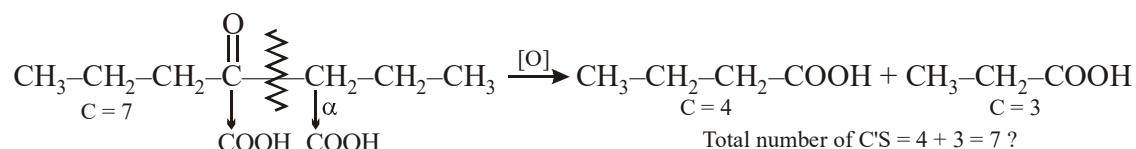


Ketones are not easy to oxidize so they do not give these 5 tests. These five tests can be used to distinguish aldehyde and ketones. Both gives 2,4 DNP test

Oxidation of ketones (with strong oxidising agent acidic KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$ etc.) : Ketones undergo oxidation only in drastic conditions.

During the oxidation of ketones there is breaking of carbon-carbon bond between α -carbon and carbonyl carbon. In this process both carbons convert into carboxylic groups. This leads to the formation of two moles of monocarboxylic acids.

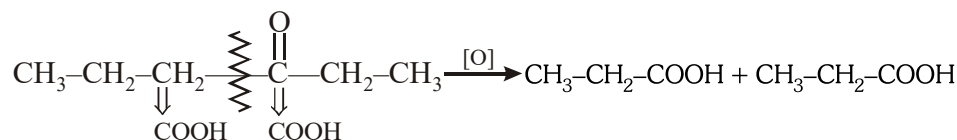
Case - I : Oxidation of Symmetrical ketones :



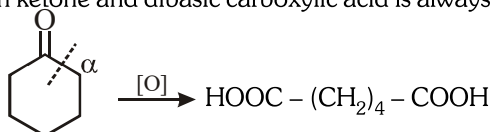
Thus number of carbons in any product is less than the number of carbons in ketone.

Case - II : Oxidation of unsymmetrical ketones (Poff's rule) : In case of unsymmetrical ketones, those

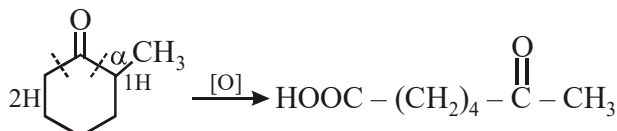
—C—R bond break in which alkyl group has more number of carbons. This rule is known as **Poff's rule**.



Case - III : Oxidation of cyclic ketones : Formation of dibasic acid takes place from cyclic ketones. In this case number of carbons in ketone and dibasic carboxylic acid is always same.

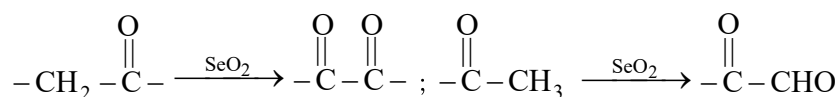


Note : If both α -carbons are not identical then ;

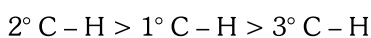
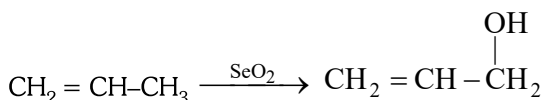


Allylic oxidation (SeO_2)

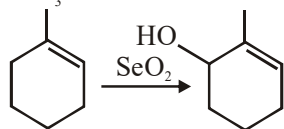
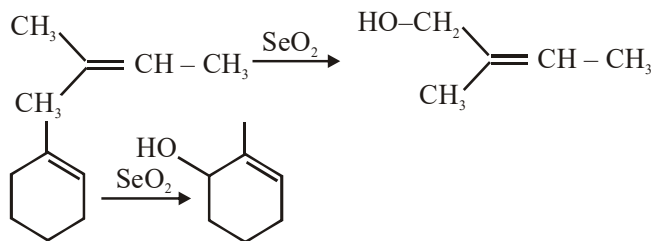
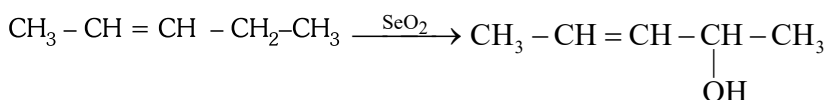
SeO_2 is a selective oxidizing agent which converts $-\text{CH}_2-$ group adjacent to carbonyl group into carbonyl group. The reagent, in general, oxidises active methylene and methyl groups to ketonic and aldehydic groups respectively.



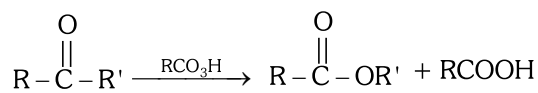
Double bonds, triple bonds and aromatic rings may also activate the methylene group. The methylene or methyl group α to the most highly substituted end of the double bond is hydroxylated according to the order of preference of oxidation $\text{CH}_2 > \text{CH}_3 > \text{CH}$ groups.



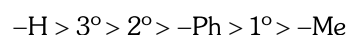
Rate of reactivity order



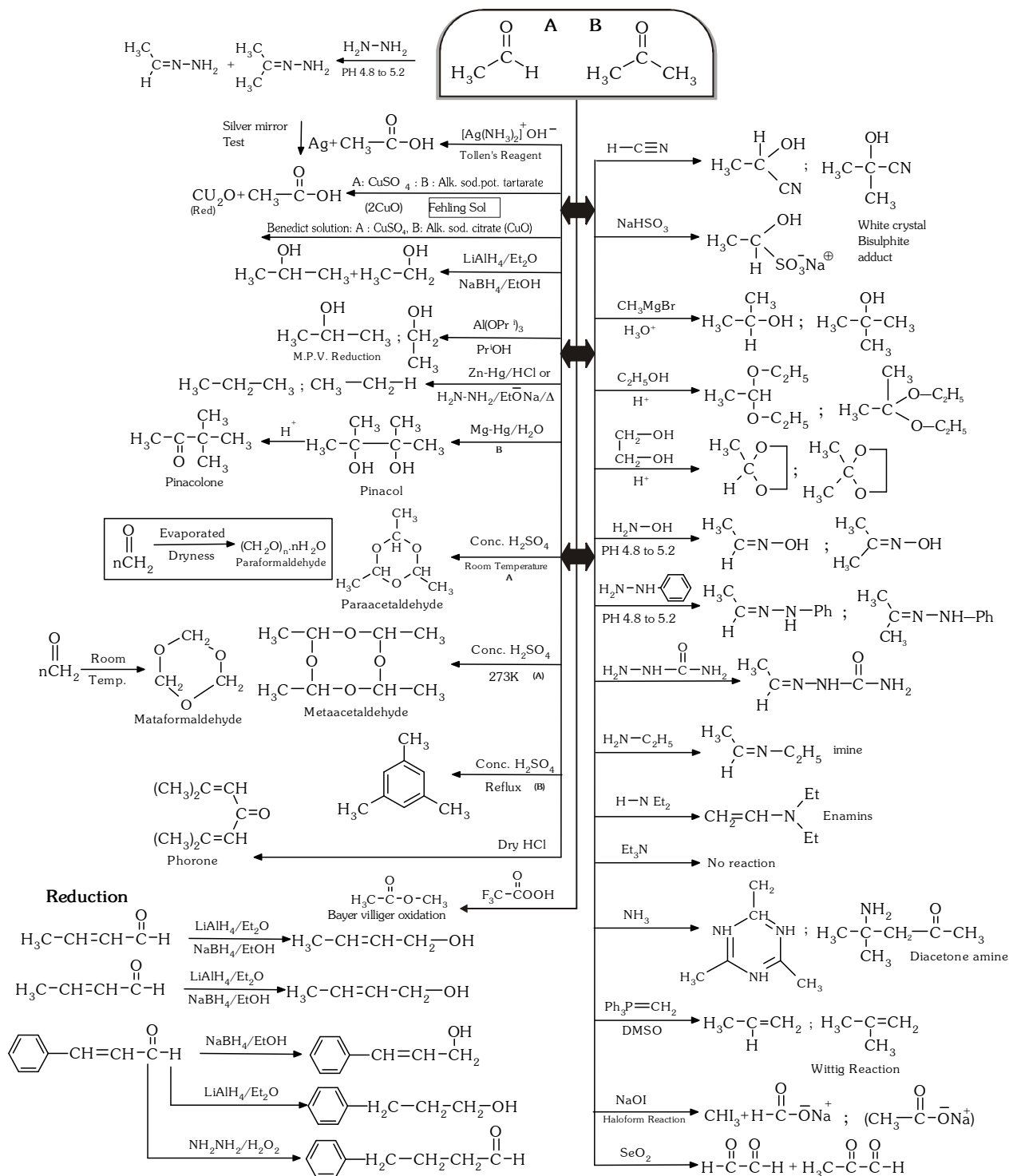
Baeyer Villiger Oxidation



Migratory order for above reaction,

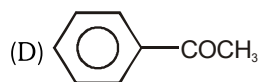
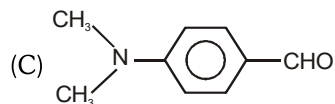
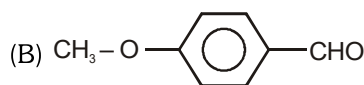
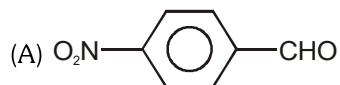


GOLDEN KEY POINTS

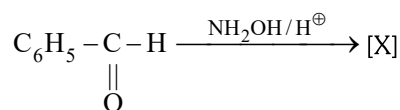


BEGINNER'S BOX-2

1. Cyanohydrin formation constant will be highest for :

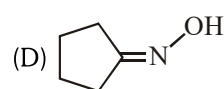
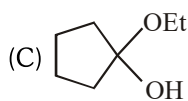
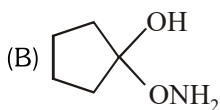
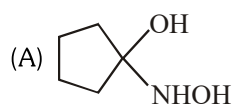
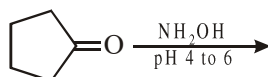


2. In the given reaction :

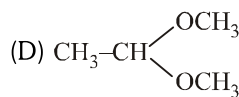
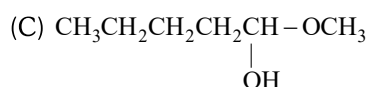
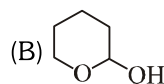
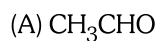


[X] will be:

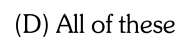
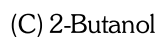
- (A) Only syn oxime (B) Only anti oxime
(C) mixture of syn and anti oxime (D) secondary amide
3. Cyanohydrin of which compound on hydrolysis will give lactic acid?
(A) $\text{C}_6\text{H}_5\text{CHO}$ (B) HCHO (C) CH_3CHO (D) $\text{CH}_3\text{CH}_2\text{CHO}$
4. Which one of the following, forms relatively more stable gem diol ?
(A) HCHO (B) CH_3CHO (C) CH_3COCH_3 (D) $\text{C}_6\text{H}_5\text{COCH}_3$
5. Which is the major product of the following reaction ?



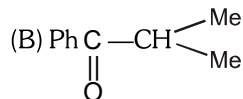
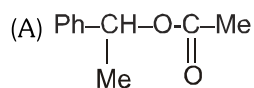
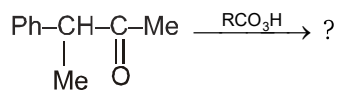
- *6. Which of the following compound will give positive Tollens test



7. Which of the following compounds is oxidised to prepare ethylmethylketone ?



8. What will be the product of the following reaction



(D) None of these

*9. Fehling solution gives red precipitate with:

(A) Aromatic aldehyde

(B) Aliphatic aldehyde

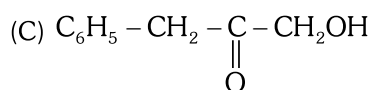
(C) Ketones

(D) α -hydroxy ketones

*10. Silver mirror test with Tollens reagent is given by :

(A) $\text{C}_6\text{H}_5\text{CHO}$

(B) $\text{Ph}-\text{NH}-\text{OH}$



(D) CH_3CHO

*11. Schiff's reagent gives pink colour with :

(A) Acetaldehyde

(B) Formic acid

(C) Acetic acid

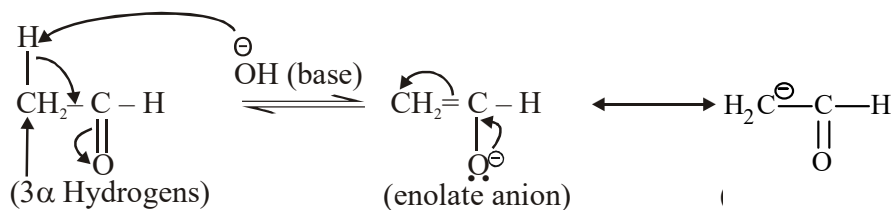
(D) Methyl acetate

ALDOL & SIMILAR KIND OF REACTION

1.0 NAMED REACTIONS

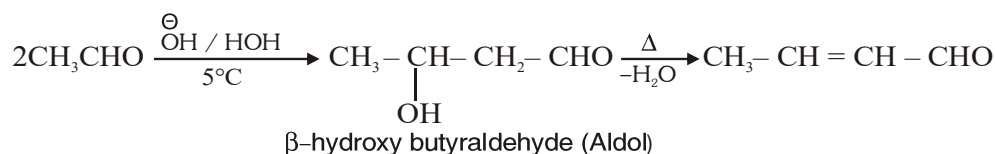
1.1 ALDOL CONDENSATION

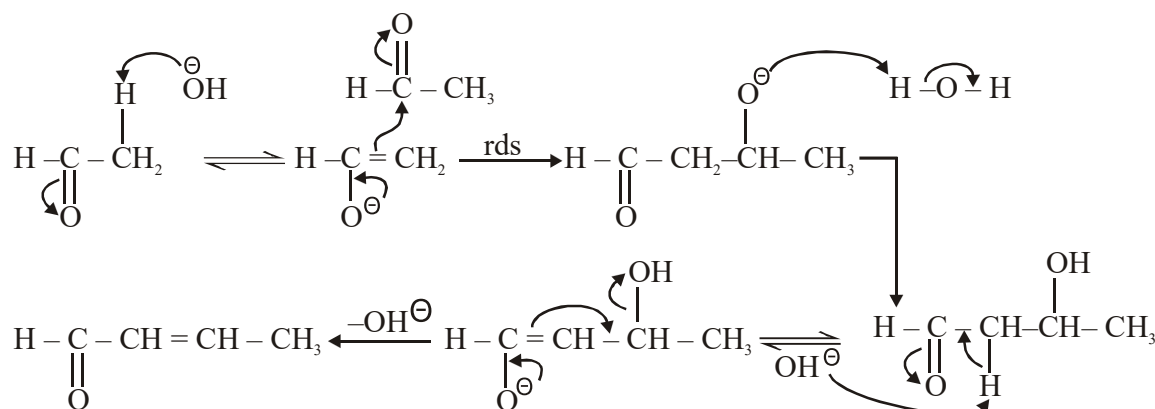
The α -hydrogen of carbonyl compounds are acidic due to the fact that the anion (enolate ion) is stabilized by resonance.



Base catalysed Aldol

In aqueous base, two acetaldehyde molecules react to form β -hydroxy aldehyde called aldol. The reaction is called Aldol condensation. The enolate ion is the intermediate in the aldol condensation of aldehyde and ketone. Acetaldehyde for instance, forms a dimeric product aldol in presence of a dilute base ($\approx 10\%$ NaOH)

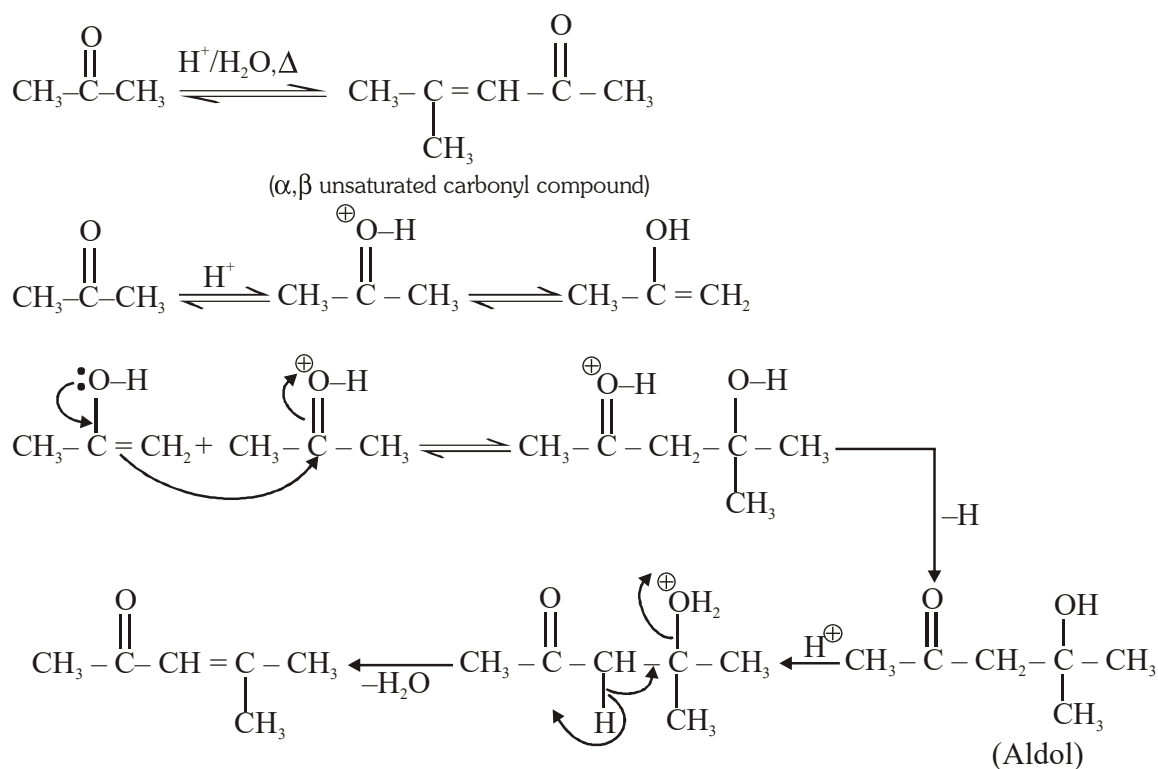


Mechanism :

Aldols are stable and may be isolated. They, however can be dehydrated easily by heating the basic reaction mixture or by a separate acid catalyzed reaction. Thus if the above reaction is heated the product is dehydrated to 2-butenal (crotonaldehyde).

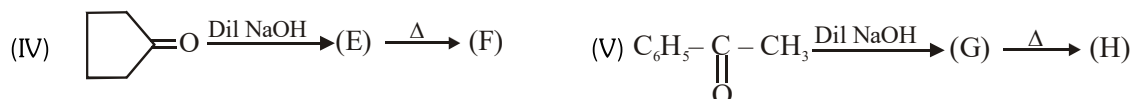
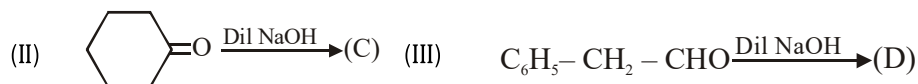
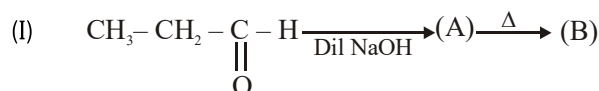
Acid catalysed Aldol

In acid catalysed aldol condensation enol form of carbonyl is the nucleophile in place of enolate.

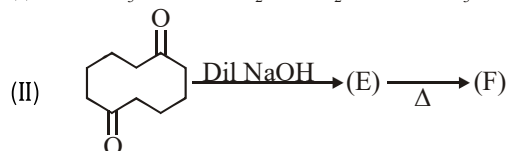
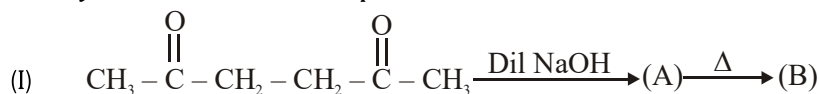
Mechanism

BEGINNER'S BOX-3

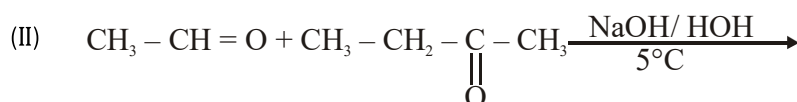
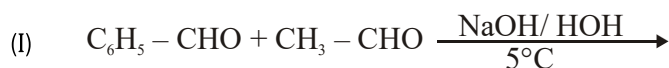
1. Write the product and mechanism for given reactions.



2. Identify the intramolecular aldol product?

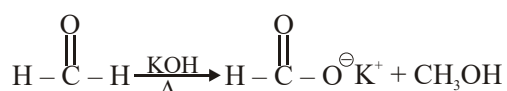


3. Find out the total number of possible aldol products (including and excluding stereo products).

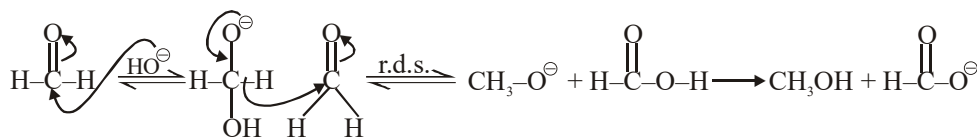


1.2 CANNIZARO REACTION

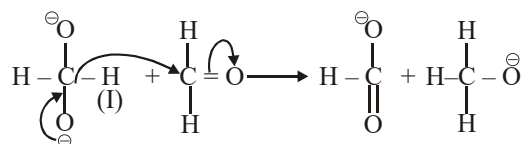
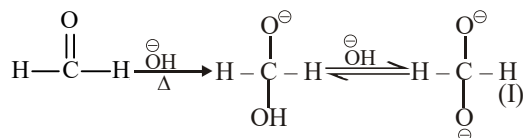
This reaction is given by aldehyde having no α -hydrogen in the presence of conc. NaOH/ Δ or KOH/ Δ .



Mechanism :

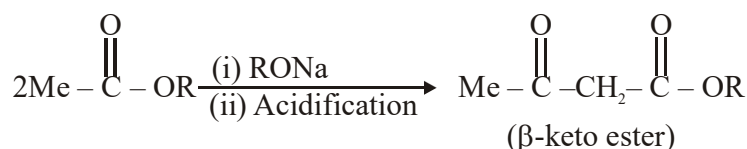


In the presence of a very strong concentration of alkali, aldehyde first forms a doubly charged anion (I) from which a hydride anion is transferred to the second molecule of the aldehyde to form acid and an alkoxide ion. Subsequently, the alkoxide ion acquires a proton from the solvent.

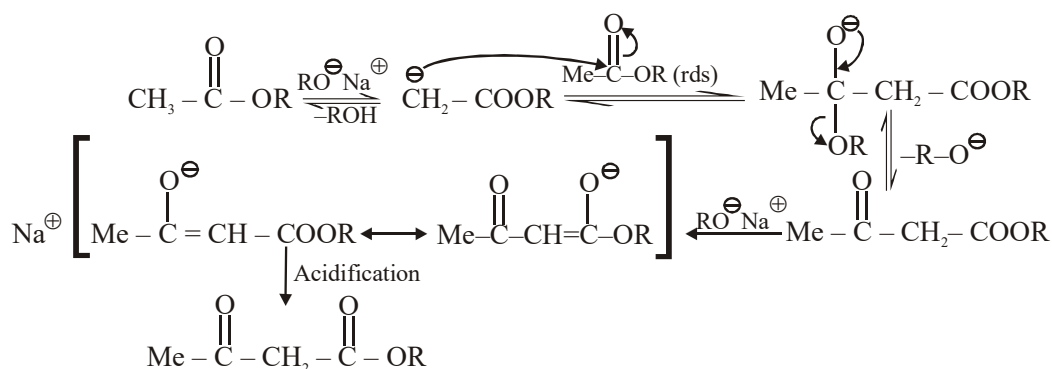


1.3 CLAISEN CONDENSATION

Esters undergo S_NAE Reaction. when attacked by a Nu^- generated by the interaction of a base (usually base related to the Alkoxy anion of ester) with one of the molecule of ester and this Nu^- attacks on another molecule. The reaction over all is considered as condensation of ester known as claisen ester condensation.



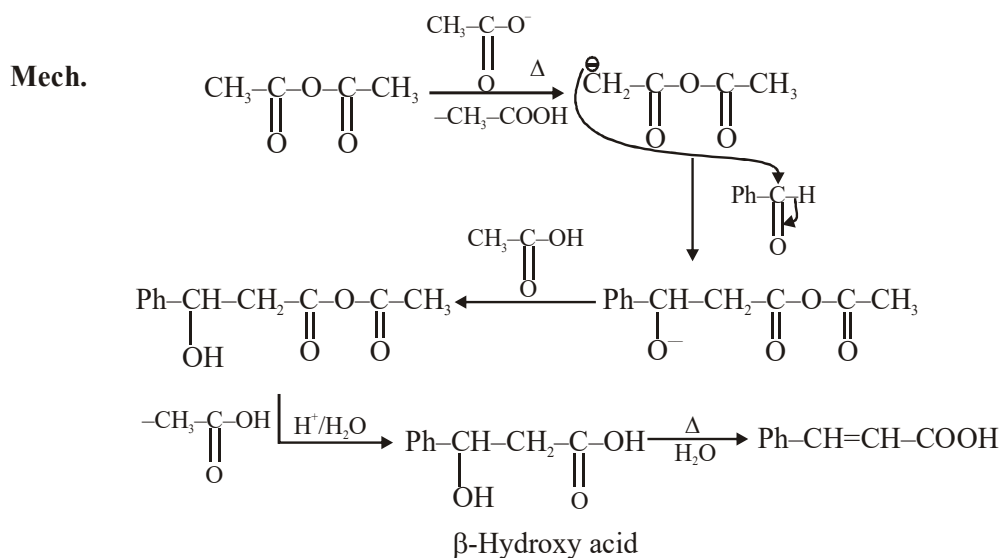
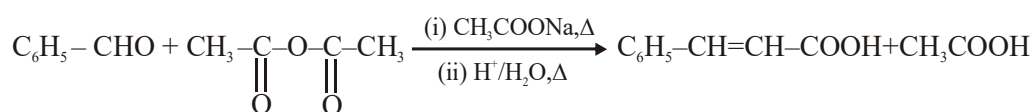
Mechanism :



Some times, when two ester group are present within the molecule then intramolecular condensation occurs and cyclization is caused, thus it is known as Dieckmann cyclization or Dieckmann's condensation.

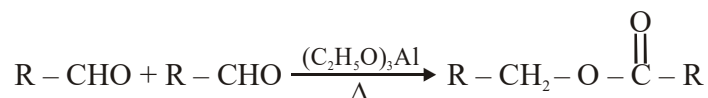
1.4 PERKIN REACTION

In Perkin reaction, condensation has been observed between aromatic aldehydes and aliphatic acid anhydrides in the presence of sodium or potassium salt of corresponding acid of that anhydride, to yield α, β - unsaturated aromatic acids.

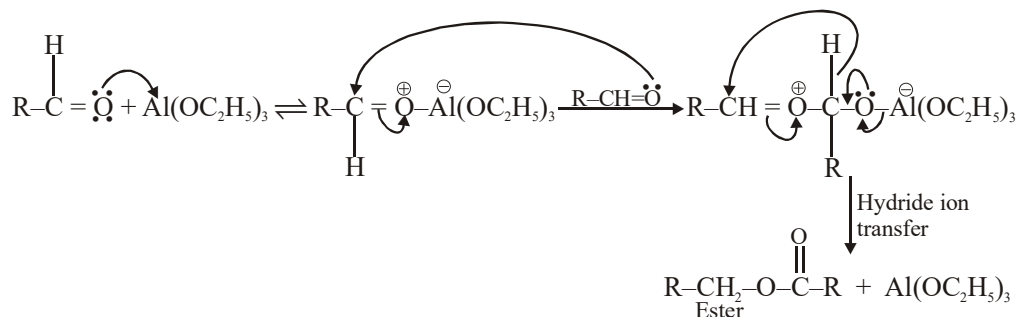


1.5 TISCHENKO REACTION /CONDENSATION

- (1) This reaction takes place between two molecules of aldehydes.
- (2) Reaction is catalysed by aluminium ethoxide.
- (3) This is a two-step reaction, i.e., redox reaction followed by ester formation. Thus this reaction is extension of Cannizzaro reaction.

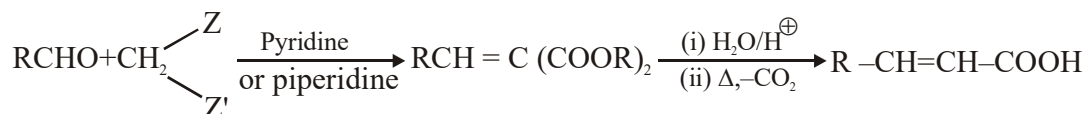


Mechanism :

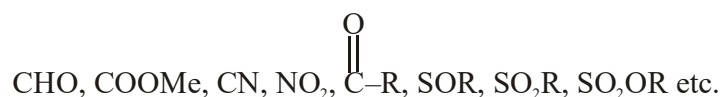


1.6 KNOEVENAGEL CONDENSATION

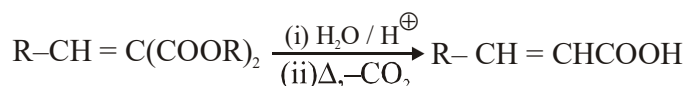
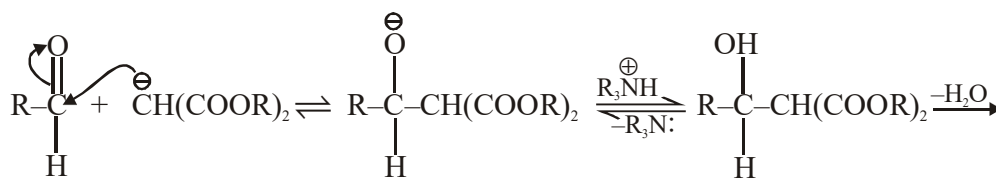
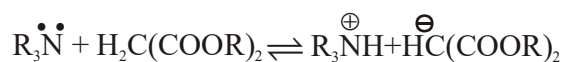
Reaction of active methylene group with aldehyde & Ketones is known as Knoevenagel reaction.



Z can be



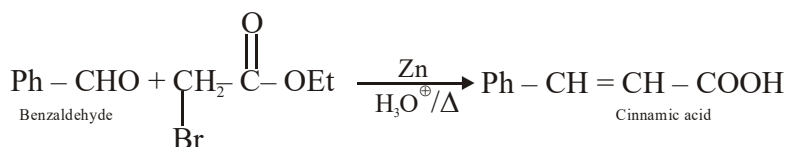
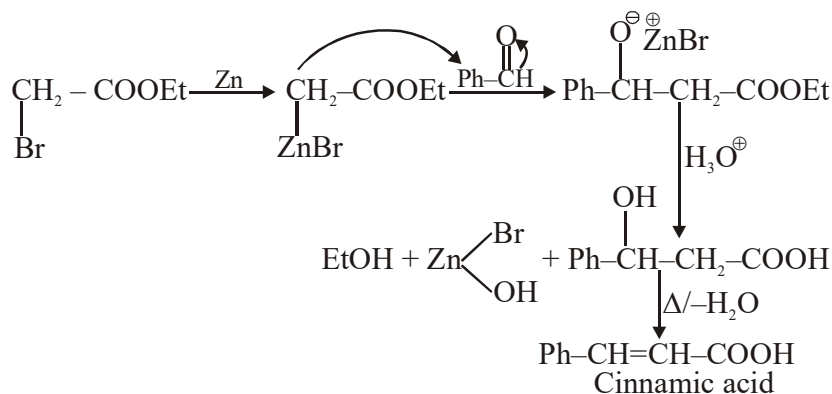
Mechanism :



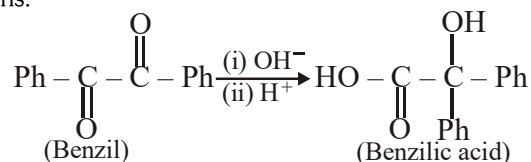
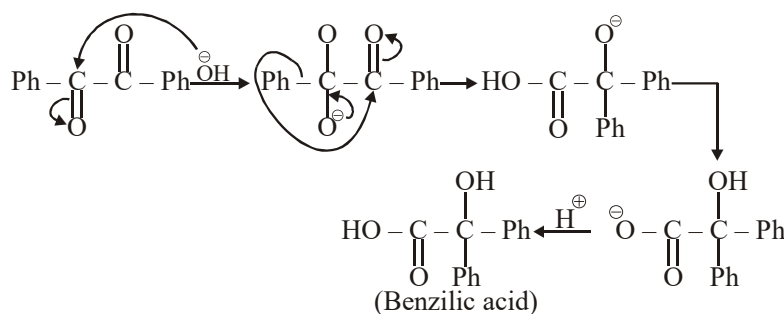
High reactivity of the methylene group of the active methylene compound prevents self-condensation of the aldehyde.

1.7 REFORMATSKY REACTION

α -halo esters when treated with Zn it gives organometallic halo ester which provides the attacking Nu^- for the another reactant, which is a carbonyl compound. When Nu^- attacks on carbonyl compound it gives an intermediate which upon acidic hydrolysis followed by heating, results in formation of α, β -unsaturated acid. The overall reaction is known as Reformatsky reaction.

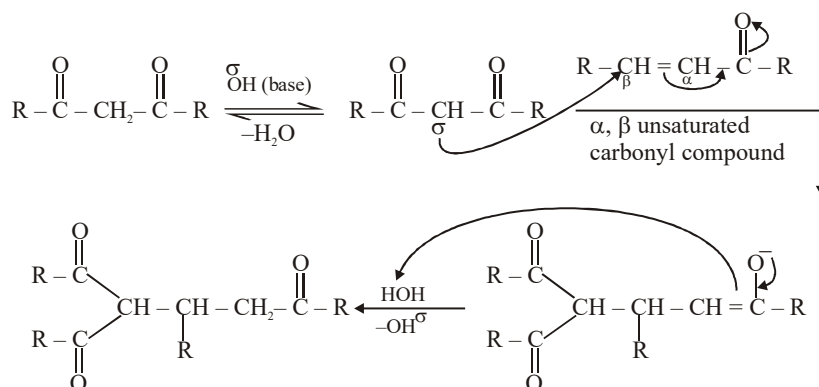
**Mechanism :****1.8 BENZIL-BENZILIC REARRANGEMENT OR BENZILIC ACID REARRANGEMENT**

The base catalysed reaction of 1,2-diketones to a salt of α -hydroxy carboxylic acid is known as Benzilic acid rearrangement, this reaction is mainly applicable when aryl group is present on both carbonyl carbons.

**Mechanism :****1.9 MICHAEL REACTION**

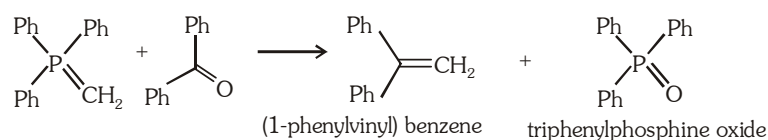
Michael Addition : α - β unsaturated carbonyl compound undergo Michael reaction with compounds having active methylene. Many different nucleophile can add to α - β unsaturated carbonyl compound. When the nucleophile is an enolate the addition reaction has a special name MICHAEL REACTION.

Mechanism :

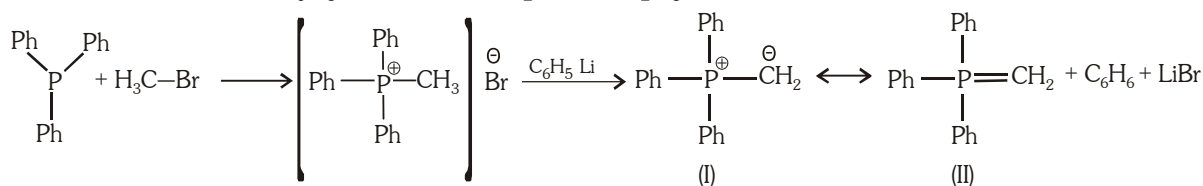


1.10 WITTING REACTION

Wittig reaction affords an important and useful method for the synthesis of alkenes by the treatment of aldehydes or ketones with alkylidenetriphenylphosphorane ($Ph_3P=CH_2$) or simply known as phosphorane

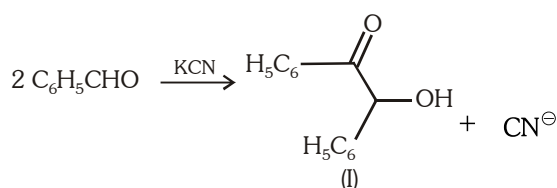


The wittig reagent, alkylidenetriphenylphosphorane (ylide), is prepared by treating trialkyl or triarylphosphine usually the latter with an alkyl halide in ether solution. The resulting phosphonium salt is treated with strong base (such as C_6H_5Li , $BuLi$, $NaNH_2$, NaH , C_2H_5ONa , etc.)



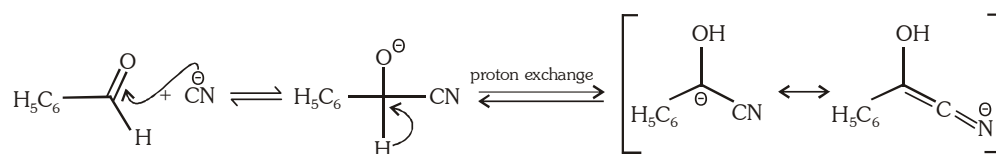
1.11 BENZOIN CONDENSATION

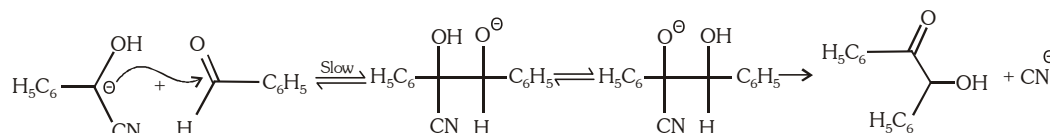
The benzoin condensation is essentially a dimerisation of two aromatic aldehydes under the catalytic influence of cyanide ions to give benzoin (I).



The hydrogen atom attached to the carbonyl group of aldehyde is not active enough to be removed easily but the addition of the cyanide ion to the carbonyl carbon places this hydrogen in the alpha position of the nitrile thus rendering it relatively acidic. The carbanion, thus generated, attacks the carbonyl carbon of the second aldehyde molecule in a rate-determining step forming an unstable cyanohydrin of benzoin which immediately breaks down into benzoin and hydrogen cyanide.

Mechanism

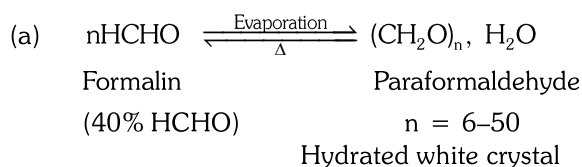




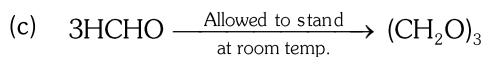
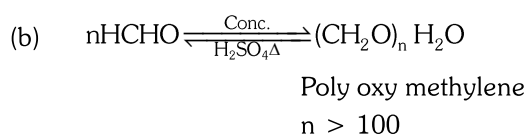
2.0 REACTIONS GIVEN BY ALDEHYDES

2.1 Polymerisation : It is a reversible process

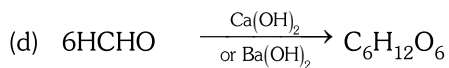
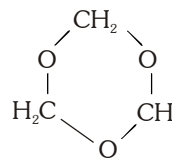
Formaldehyde :



Paraformaldehyde is a linear polymer which show reducing character with Tollen's reagent, Fehling solution etc.

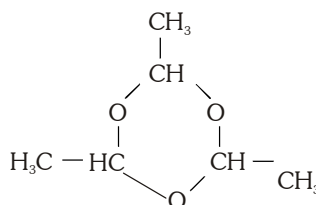


Meta formaldehyde (Trioxane)
Cyclic polymer (Trioxymethylene)
Cyclic polymer doesn't show reducing character with Tollen's reagent etc.



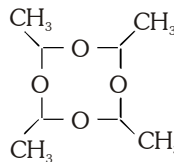
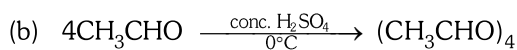
Formose sugar
A linear polymer (α -acrose)

Acetaldehyde :



Para acetaldehyde

Paraldehyde (cyclic polymer). Pleasant smelling liquid.
Hypnotic compound



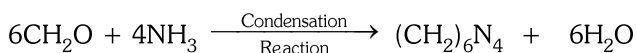
Meta aldehyd

White crystalline solid. Cyclic polymer.

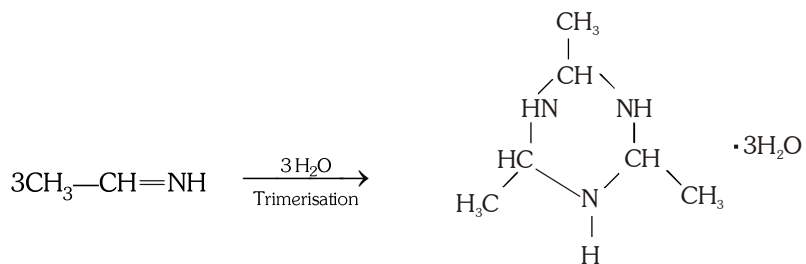
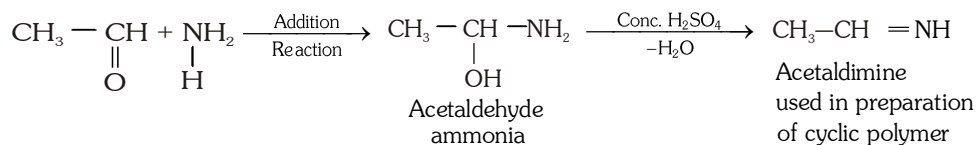
Used as solid fuel or killing snails

2.2 Reaction with ammonia

Except formaldehyde, all other aldehydes give addition reactions (HCHO give addition elimination i.e. condensation reaction)



Urotropine (Hexamethylene tetramine)
White crystalline solid. Used in preparation of explosive.
Used in treatment of urine infection diseases

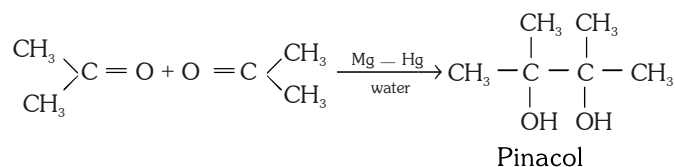


Trimethyl hexahydro Triazine trihydrate

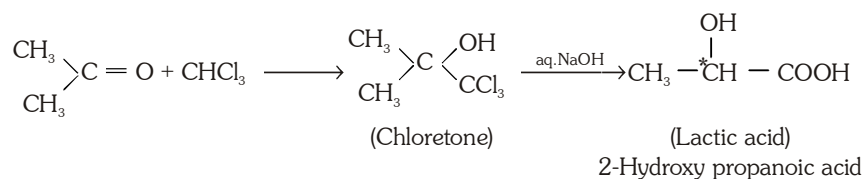
3.0 REACTION OF KETONES

3.1 Reduction

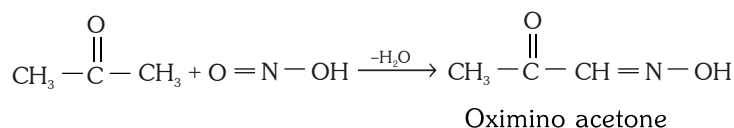
Acetone is reduced by magnesium amalgam and water to give pinacol.



3.2. Reaction with chloroform

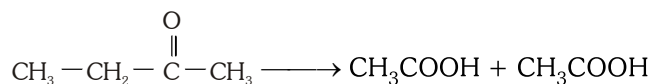


3.3 Reaction with HNO_2



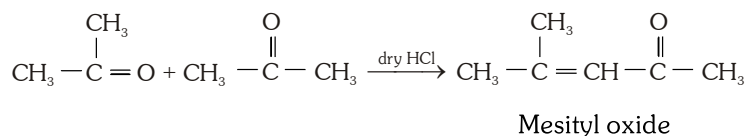
3.4 Oxidation reaction

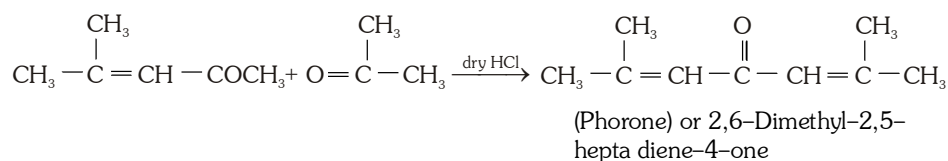
According to popoff's rule $>\text{C}=\text{O}$ group stays with smaller alkyl group.



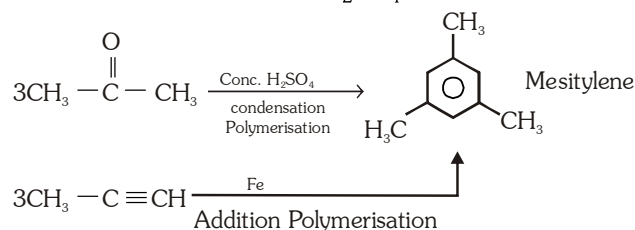
3.5 Condensation reaction

(a) In presence of dry HCl - aldol condensation takes place

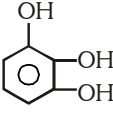




(b) In presence of conc. H_2SO_4

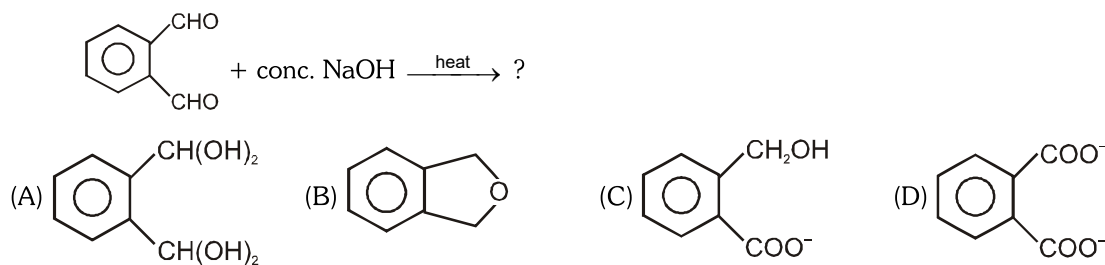


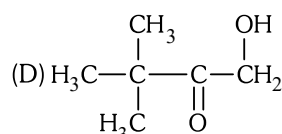
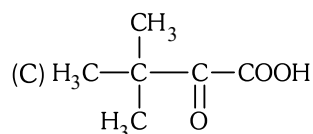
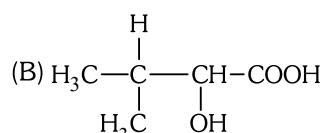
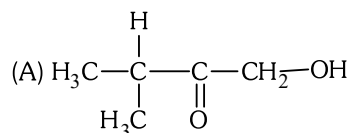
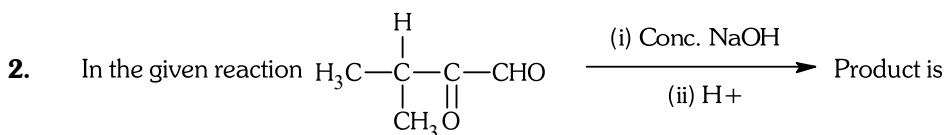
4.0 TESTS FOR HCHO , CH_3CHO , CH_3COCH_3

S.No.	Test	HCHO	CH_3CHO	CH_3COCH_3
1.	Iodoform test ($\text{I}_2 + \text{NaOH}$)	—	yellow ppt	yellow ppt
2.	Pyragallol 	white ppt.	—	—
3.	Orthonitro benzaldehyde	—	—	Blue
4.	Tollen's reagent -	Silver mirror	Silver mirror	—
	Fehling's reagent -	Red ppt	Red ppt	—
	Mercuric chloride -	Black ppt	Black ppt	—
	Schiff's reagent -	Pink colour	Pink colour	—
5.	DNP	Orange colour	Orange colour	Orange colour

BEGINNER'S BOX-4

1. Product of following reaction is

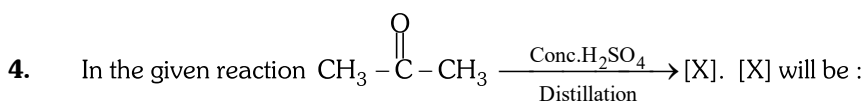




3. Product of Perkin reaction is:

- (A) α, β -unsaturated aldehyde
(C) β -Aryl- α, β -unsaturated acid

- (B) β -cyclohexyl α, β -unsaturated aldehyde
(D) All of these



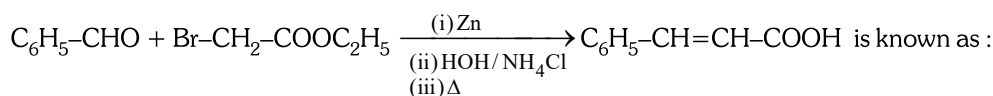
(A) Methyl oxide

(B) Phorone

(C) 1, 3, 5-Trimethylbenzene

(D) 2-Butyne

5. The given reaction

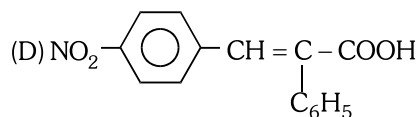
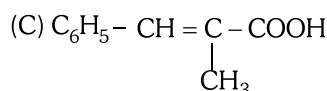
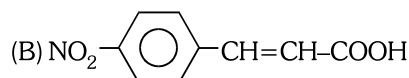
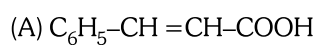
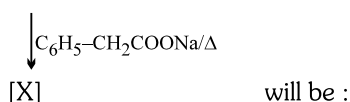
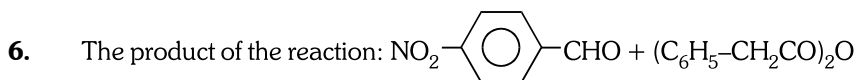


(A) Perkin reaction

(B) Knoevenagel reaction

(C) Reformatsky reaction

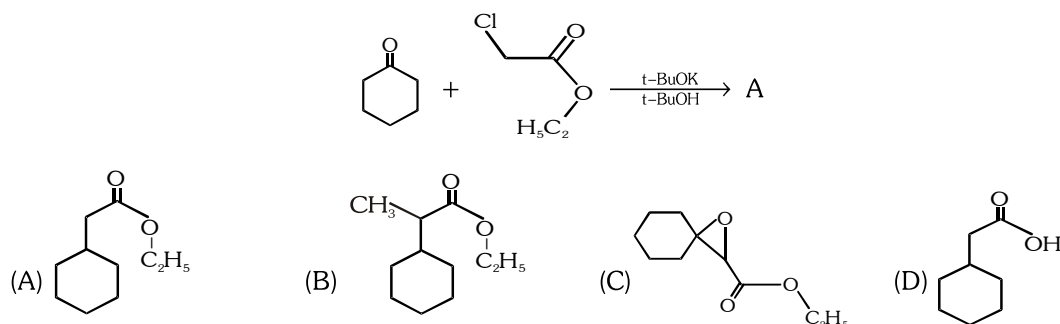
(D) Claisen-Schmidt reaction



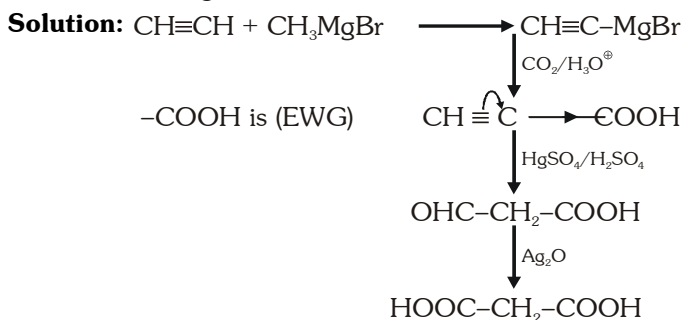
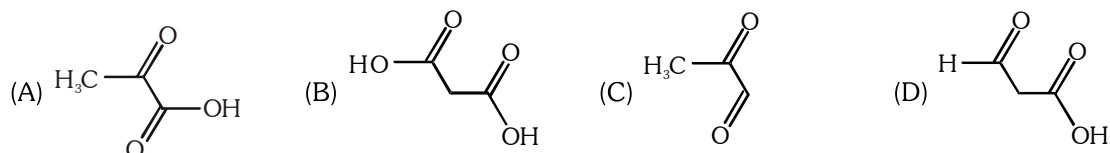
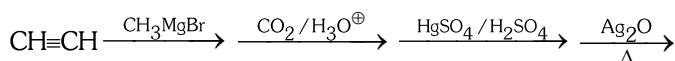
SOME WORKED OUT ILLUSTRATIONS

Illustration 1.

What is A in the following reaction ?

**Solution (C)****Illustration 2.**

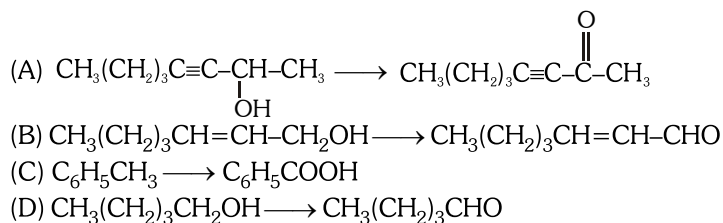
End product of the following sequence of reactions is :



$\therefore$ (B)

Illustration 3.

Which of the following oxidation reaction can be carried out with chromic acid in aqueous acetone at 5–10°C.

**Solution**

(A), (B), (D)

Illustration 4.

Product in following reaction is :

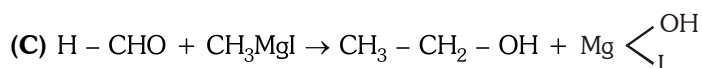
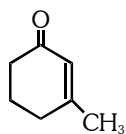
**Solution**

Illustration 5.



is the final product obtained when one of the following is reacted with base. Select the correct one.

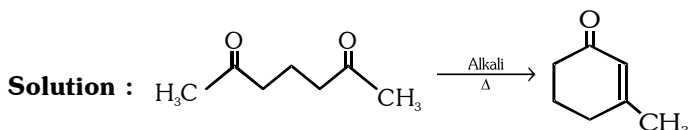
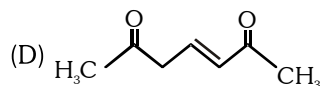
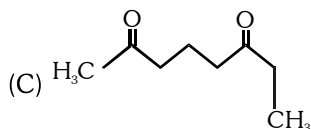
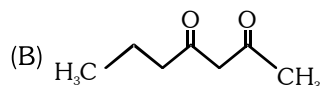
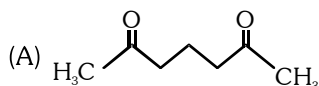
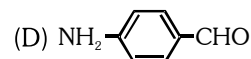
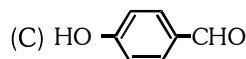
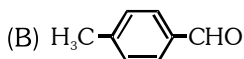
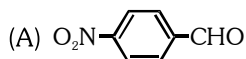
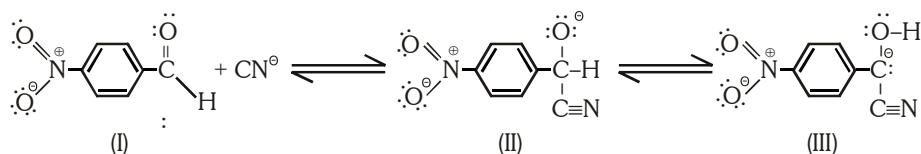


Illustration 6.

In which of the following substrates, rate of Benzoin condensation will be maximum ?



Solution



Benzoin condensation is due to stability of intermediate (III) when negative charge on C extensively delocalised in benzene ring, nitro and C≡N group. In all other cases, such dispersal is not extensively possible. On the other hand, NO₂⁻ is also creating a positive charge center on carbonyl carbon, making it more susceptible to nucleophilic attack of CN⁻

Illustration 7.

This intermediate is converted into product in the wittig reaction :

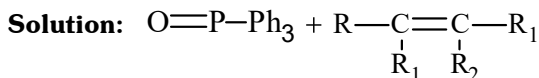
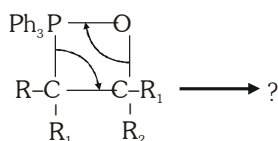
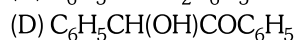
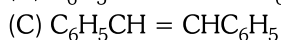
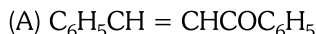


Illustration 4.

Benzaldehyde on reaction with acetophenone in the presence of sodium hydroxide solution gives :



Solution

